

Day-2: File Ownership and Access Control

Topics: chown, chgrp, ownership + permissions

Part A — Ownership Investigation (10 Minutes)

- Create a directory named ownership_lab.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ mkdir ownership_lab
ubuntu@ubuntu-VMware-Virtual-Platform:~$ cd ownership_lab
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$
```

- Inside it, create:

incident_report.txt

network.log

credentials.txt

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ touch incident_report.txt network.log credentials.txt
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$
```

- Display the detailed information for these files.

For each file, identify:

Owner

Group

Owner permissions

Group permissions

Others permissions

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ ls -l
total 0
-rw-rw-r-- 1 ubuntu ubuntu 0 Sep  3 09:41 credentials.txt
-rw-rw-r-- 1 ubuntu ubuntu 0 Sep  3 09:41 incident_report.txt
-rw-rw-r-- 1 ubuntu ubuntu 0 Sep  3 09:41 network.log
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$
```

- Determine the UID and GID of the current user.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ id
uid=1000(ubuntu) gid=1000(ubuntu) groups=1000(ubuntu),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users),111(lpadmin),114(lxd)
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$
```

- Compare the UID/GID information with the ownership displayed for the newly created files.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ ls -ln
total 0
-rw-rw-r-- 1 1000 1000 0 Sep  3 09:41 credentials.txt
-rw-rw-r-- 1 1000 1000 0 Sep  3 09:41 incident_report.txt
-rw-rw-r-- 1 1000 1000 0 Sep  3 09:41 network.log
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$
```

- Explain the difference between file ownership and file permissions.
 - File ownership: Ownership tells us who owns the file and which group is associated with it.
 - File permissions: Permissions determine what users are allowed to do.

So ownership identifies the users/groups associated with the file, while permissions determine what those users/groups can do with the file.

Part B — chown: Changing Ownership (15 Minutes)

- Change the owner of incident_report.txt to alice.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ sudo chown alice incident_report.txt
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$
```

- Change the owner of network.log to bob.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ sudo chown bob network.log
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$
```

- Change the owner of credentials.txt to charlie.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ sudo chown bob network.log
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$
```

- Verify each change.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ ls -l incident_report.txt
-rw-rw-r-- 1 alice ubuntu 0 Sep  3 09:41 incident_report.txt
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ ls -l network.log
-rw-rw-r-- 1 bob ubuntu 0 Sep  3 09:41 network.log
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ ls -l credentials.txt
-rw-rw-r-- 1 ubuntu ubuntu 0 Sep  3 09:41 credentials.txt
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$
```

- Try changing the owner of a file to another user without using sudo. Record the result.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ chown alice network.log
chown: changing ownership of 'network.log': Operation not permitted (os error 1)
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$
```

- Explain why an ordinary user is generally not permitted to transfer file ownership to another user.

An ordinary user generally cannot transfer ownership of a file to another user because that could allow users to manipulate ownership and potentially bypass administrative controls. Normally, changing the owner requires root privileges.

- Using a single command, configure:

incident_report.txt

Owner : alice

Group : security

Verify the resulting ownership.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ sudo chown alice:security incident_report.txt
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ ls -l incident_report.txt
-rw-rw-r-- 1 alice security 0 Sep  3 09:41 incident_report.txt
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$
```

Part C — chgrp: Changing Group Ownership (15 Minutes)

- Change the group ownership of network.log to developers.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ sudo chgrp developers network.log
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$
```

- Change the group ownership of credentials.txt to security.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ sudo chgrp security credentials.txt
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$
```

- Verify the changes.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ ls -l network.log
-rw-rw-r-- 1 bob developers 0 Sep  3 09:41 network.log
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ ls -l credentials.txt
-rw-rw-r-- 1 ubuntu security 0 Sep  3 09:41 credentials.txt
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$
```

- Display the group memberships of alice, bob, and charlie.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ groups alice
alice : alice security
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ groups bob
bob : bob security developers
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ groups charlie
charlie : charlie security
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$
```

- Log in as alice and create a file named alice_report.txt.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ sudo passwd alice
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ su - alice
Password:
$ whoami
alice
$ touch alice_report.txt
$
```

- Attempt to change the group of alice_report.txt to security.

```
$ chgrp security alice_report.txt
$
```


- Attempt to change its group to developers.

```
$ chgrp developers alice_report.txt
chgrp: changing group of 'alice_report.txt': Operation not permitted (os error 1)
$
```

- Compare the results.

Alice was able to change the group ownership to security because she is a member of that group. She could not change it to developers because she is not a member of that group. Thus, a user's group membership directly affects which groups they can assign to files they own.

- Explain how a user's group membership affects their ability to change the group ownership of a file they own.

A user's group membership determines which group ownerships they are allowed to assign to a file they own.

- A normal user who owns a file can generally change its group only to a group that the user is a member of.
- If the user is not a member of the target group, the chgrp command normally fails with “Operation not permitted.”
- For example, if Alice owns alice_report.txt and belongs to security, she can change its group to security.
- If Alice does not belong to developers, she generally cannot change the file's group to developers.
- Root/sudo can change the group ownership regardless of the user's group memberships.

Part D — Ownership + Permission Investigation (20 Minutes)

- Configure:

-rw-r----- alice security incident_report.txt

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ sudo chown alice:security incident_report.txt
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ sudo chmod 640 incident_report.txt
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ ls -l incident_report.txt
-rw-r----- 1 alice security 0 Sep  3 09:41 incident_report.txt
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$
```

- Identify what operations are available to:

Alice: Can read and modify the file. (Owner)

Charlie: Can only read the file. (Security)

Bob: Can only read the file. (Security)

A user who belongs to neither security nor the file owner

charlie belongs to security.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ groups charlie
charlie : charlie security
```

- Test whether Charlie can:

Read incident_report.txt: can read

Modify incident_report.txt: cannot modify

Delete incident_report.txt: cannot delete

Record the result of each operation.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ su - charlie
Password:
$ cd /home/ubuntu/ownership_lab
$ ls -l
total 0
-rw-rw-r-- 1 ubuntu security  0 Sep  3 09:41 credentials.txt
-rw-r----- 1 alice  security  0 Sep  3 09:41 incident_report.txt
-rw-rw-r-- 1 bob    developers 0 Sep  3 09:41 network.log
$ cat incident_report.txt
$ echo "Charlie test" >> incident_report.txt
-sh: 4: cannot create incident_report.txt: Permission denied
$ rm incident_report.txt
rm: remove write-protected regular empty file 'incident_report.txt'? y
rm: cannot remove 'incident_report.txt': Permission denied
$
```

- Change the permissions to:

-rw-rw----

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ sudo chmod 660 incident_report.txt
[sudo: authenticate] Password:
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ ls -l incident_report.txt
-rw-rw---- 1 alice security 0 Sep  3 09:41 incident_report.txt
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$
```

- Repeat the test as charlie.

Read incident_report.txt: can read

Modify incident_report.txt: can modify

Delete incident_report.txt: cannot delete

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ su - charlie
Password:
$ cd /home/ubuntu/ownership_lab
$ cat incident_report.txt
$ echo "Charlie test" >> incident_report.txt
$ rm incident_report.txt
rm: cannot remove 'incident_report.txt': Permission denied
$ exit
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$
```

- Explain why Charlie's access changed even though the owner and group of the file did not change.

Charlie's access changed because the group permission was changed from r-- to rw-. Charlie is a member of the security group, so he could read the file before, but after the permission change he could also modify it. The owner and group of the file remained the same.

- Remove bob from the security group.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ groups bob
bob : bob security developers
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ sudo gpasswd -d bob security
Removing user bob from group security
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ groups bob
bob : bob developers
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$
```

- Without changing the file's ownership or permissions, test Bob's access again.

Read incident_report.txt: cannot read

Modify incident_report.txt: cannot modify

Delete incident_report.txt: cannot delete

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$ su - bob
Password:
$ cd /home/ubuntu/ownership_lab
$ cat incident_report.txt
cat: incident_report.txt: Permission denied
$ echo "Bob test" >> incident_report.txt
-sh: 3: cannot create incident_report.txt: Permission denied
$ rm incident_report.txt
rm: remove write-protected regular file 'incident_report.txt'? y
rm: cannot remove 'incident_report.txt': Permission denied
$ exit
ubuntu@ubuntu-VMware-Virtual-Platform:~/ownership_lab$
```

- Explain why Bob's effective permissions changed.

Bob's effective permissions changed because he was removed from the security group. Although the file still has the same owner, group, and permissions, Bob is no longer a member of the group. Therefore, he receives the others permissions (---) instead of the group's rw- permissions.